A Survey of Optimization Approaches for Wireless Physical Layer Security

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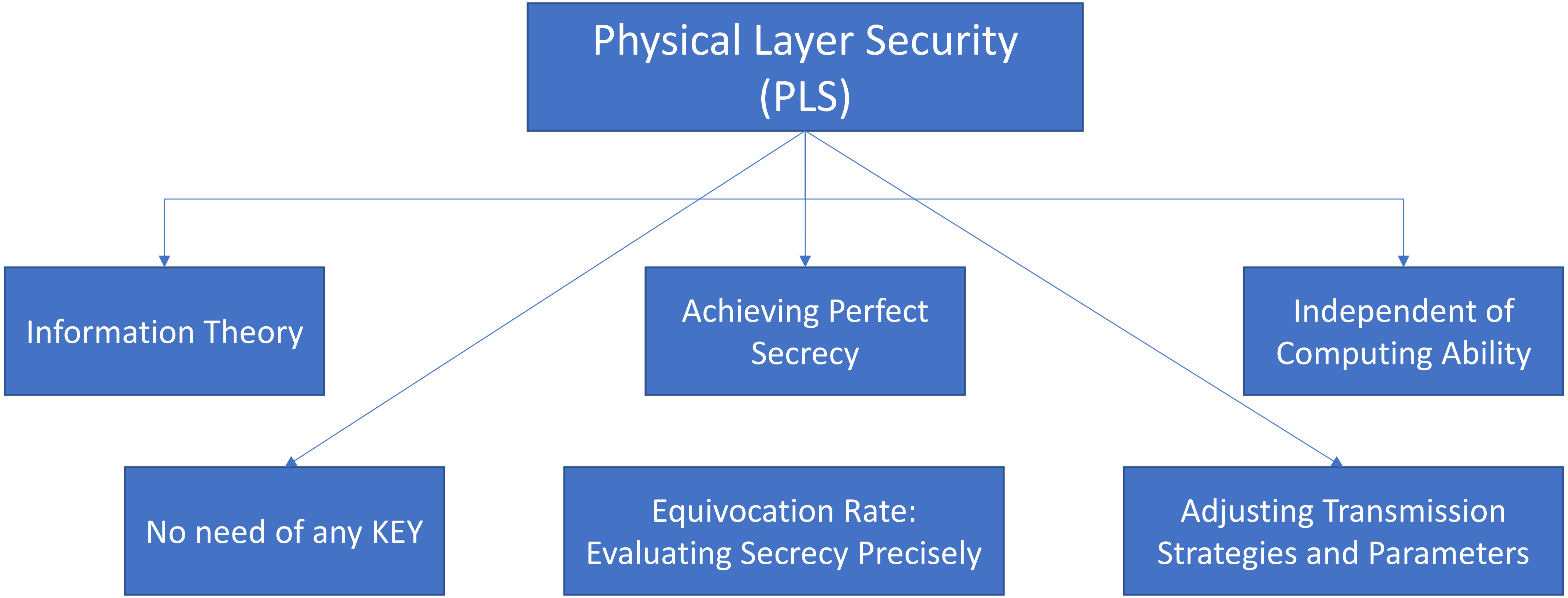
Cryptographic Encryption Vs. Physical Layer Security



System Designs of Physical Layer Security

- Secure Resource Allocation
  - Increase the difference between the legitimate channel and the wiretap channel by secure resource allocation.
  - Subcarrier allocation (binary integer programming)
    - decision variable α ∈{0, 1}, with α =1 meaning that the subcarrier is used for transmitting and α =0 otherwise.
  - Power allocation
  - Joint Subcarrier and Power allocation (mixed integer nonlinear optimization)
- Secure Beamforming and Precoding
  - Beamforming: Single data stream is transmitted over multiple antennas or nodes
    - compute the optimal beamforming vector (Enhance Signal quality at Legitimate User, Decrease at Eavesdropper)
    - Null-space beamforming: chooses the beamforming vector lying in the null space of the eavesdropper's channel vector.
    - Masked beamforming: Uses Artificial Noise (AN) to reduce the quality of signal at Eaves
    - to avoid interfering destination node, the simple but not optimal method is to let the AN lie in the **null space** of the signal space
  - Precoding: more than one data streams can be transmitted at the same time
    - the confidential messages of one or multiple users can be spatially multiplexed onto multiple independent subchannels via precoding.

Cont.

- Antenna/Node Selection and Cooperation
  - Antenna Selection
    - exploit spatial degrees of freedom in multi-antenna scenarios,
    - maximizing the secrecy rate,
    - improving the SNR of the legitimate channels, and
    - enhancing security from the viewpoint of secrecy outage performance
  - User Selection
  - Relay Node Selection
  - Joint Relay and Jammer Selection
- Joint Strategies of Several Approaches
  - Joint **resource allocation** and **user scheduling**
  - **Antenna selection** combined with **beamforming/precoding**
  - **Distributed beamforming** with **relay/jammer selection**
  - Joint **power allocation** and **beamforming/precoding**
  - **Jamming-**aided **beamforming/precoding**
  - cooperative **beamforming** and **user selection**

Performance Metrics

- Choice of performance metrics
  - Achievable secrecy rate/capacity (transmission effectiveness)
  - Secrecy outage probability/capacity (reliability)
    - the probability that a secrecy outage event happens.
    - the secrecy outage happens when the instantaneous secrecy capacity C_s drops below a target secrecy rate R_0
    - secrecy outage capacity: the largest secrecy rate that can be supported under a tolerable secrecy outage probability
  - Minimum power consumption needed (power cost)
  - Energy Efficiency of secure transmission strategies (energy consumption)
    - the amount of secret bits transmitted with unit energy consumption
    - the ratio of secrecy rate to total power consumption

Extending Optimization Problems to System Designs

1. Secure Resource Allocation

I. Maximization of Achievable Secrecy Rate:

- Conventional Approach:
  - Globally Allocate The Limited Power And Subcarriers For All Transmission Nodes
  - Mixed Integer Programming
  - decision variables $\mu_{si} \in \{0, 1\}$ and $\mu_{ri} \in \{0, 1\}$ are defined for the source and the relay, respectively, for specifying the state of communication on a carrier i.
    - if $\mu_{si} = 1$ and $\mu_{ri} = 1$ then both the source and the relay transmit in respective slots
    - if $\mu_{si} = 1$ and $\mu_{ri} = 0$ then only the source transmits in its slot and it remains silent with the relay in the second slot.
    - if $\mu_{si} = 0$ and $\mu_{ri} = 0$: indicates no communication in both the slots
    - if $\mu_{si} = 0$ and $\mu_{ri} = 1$: no significance
  - The resource allocation strategy for maximizing secrecy capacity in such a **relay-aided multicarrier system** can be derived by solving the typical mixed integral programming

$$\max_{P_{si}, P_{ri}, \mu_{si}, \mu_{ri}} \sum_i C_i(P_{si}, P_{ri}, \mu_{si}, \mu_{ri})$$

$$\text{s.t.} \quad \begin{cases} \sum_i \mu_{si}(P_{si} + \mu_{ri} P_{ri}) \leq P_{sum}^{max} \\ P_{si} \geq 0, P_{ri} \geq 0 \\ \mu_{si} \in \{0, 1\}, \mu_{ri} \in \{0, 1\}, \end{cases}$$

where P_{si}, P_{ri}, and C_i denote the source power, the relay power, and the secrecy capacity on carrier i, respectively.

Cont.

II. Minimization of Secrecy Outage Probability

- **Secure Resource Allocation** is also an effective approach for minimizing secrecy outage probability.
- outage-optimal subcarrier allocation:
  - minimize the secrecy outage probability p^m_{out} of each user m while guaranteeing that
  - each user has the identical probability to access each subcarrier n.

The formulation of the problem is summarized as

$$\min \ \{p^m_{out}\}_{\forall m}$$
$$\text{s.t.} \begin{cases} \sum_{\forall m} \mu_{mn} \leq 1 \\ \sum_{\forall n} \mu_{mn} \leq 1 \\ \mu_{mn} \in \{0, 1\}, \end{cases} \quad (38)$$

where μ_{mn} are the decision variables with $\mu_{mn} = 1$ meaning that subcarrier n is assigned to user m. Otherwise, $\mu_{mn} = 0$. The constraints $\sum_{\forall m} \mu_{mn} \leq 1$ and $\sum_{\forall n} \mu_{mn} \leq 1$ imply that each subcarrier can only be assigned to no more than one user with identical probability.

Cont.

III. Minimization of Power Consumption:

- The power consumption of physical-layer secure communications can also be decreased by the designs of **Secure Resource Allocation.**
  - The use of AN or jamming signals can weaken the quality of wiretap channel
  - but it also increases the total power consumption
  - Optimal power allocation between the **desired information** and **AN/jamming** signals

IV. Maximization of Secure EE

- Multi-antenna networks, Relay networks, Cognitive radio networks
  - optimization based on fractional programming, sequential convex optimization, penalty function method and dual decomposition.
- The trade-off between energy and secrecy

Cont.

2. Secure Beamforming

- Beneficial in terms of secrecy rate, secrecy outage probability, power consumption, and secure EE

I. Maximization of Achievable Secrecy Rate:

- **If eavesdropper's CSI is unavailable**, a **joint scheme** with MRT (maximum ratio transmission) and AN (Artificial Noise) null-space beamforming
- The MRT beamforming controls the beam towards the intended user for strengthening its received signals.
- MRT beamforming may lead to information leakage on the direction to the eavesdropper,
- the AN null-space beamforming can then be exploited to disrupt the reception at the eavesdropper by emitting AN on the null space of legitimate channels.
- **If full CSI of the eavesdropper available**, the **ZF beamforming** can then be performed to overcome the information leakage to the eavesdropper by completely suppressing the beam towards the eavesdropper

II. Minimization of Secrecy Outage Probability

- AN-assisted beamforming is performed for degrading the eavesdroppers' channels
- the optimal power allocation between the confidential information and AN is obtained in closed form to minimize the secrecy rate outage probability
- single-stream beamforming (based on MRT and ZF beamforming) and the use of AN in the null space of the legitimate channels
- location-based beamforming is optimally designed to minimize the secrecy outage probability in Rician wiretap channels

Cont.

III. Minimization of Power Consumption:

- secure beamforming and precoding are also used to support the designs of **power minimization** in different scenarios.
- By optimizing beamformer of the relays, power allocated for transmitting confidential information is minimized, so that as much power as possible can be used to transmit **AN** to confuse the eavesdropper
- The problems of power minimization by beamforming/precoding are discussed under Multi-antenna broadcast networks, Distributed antenna systems, Multi-cell multigroup multicast systems and Cognitive radio networks.

IV. Maximization of Secure EE

Wiretap Channel Models

	MIMO Wiretap Channels	**Broadcast Wiretap Channels**	**Multiple-Access Wiretap Channels**	**Interference Wiretap Channels**	**Relay Wiretap Channels**
Channel Models	MISO, SIMO, MIMO	Compound Wiretap Channel	SISO, MAC, Common and Confidential Messages	SISO Interference Channel, Interference Channel with Separate Confidential Messages	Decode-and-Forward, Amplify-and-Forward
No. of Antennas	Multiple	Multiple	Multiple	Multiple	Multiple
Transmitter	1	1	Multiple	Multiple	Multiple
Receiver	1	Multiple	1	Multiple	2 (destination and relay)
Eavesdropper	1	Multiple	1	1	1

THANK YOU